

Algorithms & Complexity: Homework #1

Due on January 15, 2025

Professor Isaac Newton Section A

FirstName LASTNAME

Problem 1

Give an appropriate positive constant c such that $f(n) \leq c \cdot g(n)$ for all $n > 1$.

1. $f(n) = n^2 + n + 1, g(n) = 2n^3$
2. $f(n) = n\sqrt{n} + n^2, g(n) = n^2$
3. $f(n) = n^2 - n + 1, g(n) = n^2/2$

Solution

We solve each algebraically to find a valid constant c .

Part One

$$n^2 + n + 1 \leq n^2 + n^2 + n^2 = 3n^2 \leq c \cdot 2n^3$$

Thus $c = 2$ is valid.

Part Two

$$n^2 + n\sqrt{n} = n^2 + n^{3/2} \leq 2n^2 \leq c \cdot n^2$$

Thus $c = 2$.

Part Three

$$n^2 - n + 1 \leq n^2 \leq c \cdot n^2/2$$

Thus $c = 2$.

Problem 2

Let $\Sigma = \{0, 1\}$. Construct a DFA A that recognizes binary numbers divisible by 5.

Let state q_k represent the remainder $k \bmod 5$.

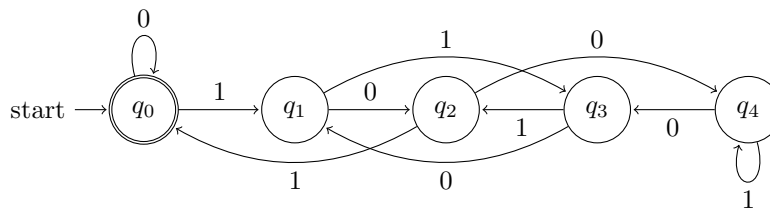


Figure 1: DFA recognizing binary multiples of 5.

Problem 3

Write the pseudocode for **Quick-Sort**($list, start, end$).

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1: function QUICK-SORT( $list, start, end$ )
2:   if  $start \geq end$  then
3:     return
4:   end if
5:    $mid \leftarrow \text{PARTITION}(list, start, end)$ 
6:   QUICK-SORT( $list, start, mid - 1$ )
7:   QUICK-SORT( $list, mid + 1, end$ )
8: end function
  
```

Algorithm 1: Quick-Sort

Problem 4

Fit a line through the origin: $Y_i = \beta_1 x_i + e_i$ with $E[e_i] = 0$ and $\text{Var}[e_i] = \sigma^2$.

Part A

Find the least squares estimator $\hat{\beta}_1$.

Solution

Minimising RSS gives:

$$\hat{\beta}_1 = \frac{\sum x_i Y_i}{\sum x_i^2}$$

Part B

Compute the bias and variance of $\hat{\beta}_1$.

Solution

$$E[\hat{\beta}_1] = \beta_1 \text{ (unbiased)}, \quad \text{Var}[\hat{\beta}_1] = \frac{\sigma^2}{\sum x_i^2}.$$